

Tucson Blooms with Less Water

Water is an increasingly precious resource that needs to be used more wisely, and there are energy penalties associated with heating and distributing water inefficiently (see “Are You Getting into Hot Water?” *HE* Sept/Oct '03, p. 33). Using less water in the home and for landscaping outside the home lessens the energy burden on water treatment and pumping



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plants. The 818-acre, mixed-use Community of Civano, in the hot-dry climate of Tucson, Arizona, was built with both energy and water saving in mind (see “Civano: Green Development at Work,” *HE* Sept/Oct '98, p. 21). But a recent meeting of Civano residents revealed some challenges associated with one water-saving measure—using reclaimed water for landscaping purposes.

Civano was built according to the Tucson Metropolitan Energy Commission's Sustainable Energy Standards (SES), which require in new homes a reduction of 50% in energy used for heating and cooling compared to homes built according to the 1995 Model Engineering Code (MEC). Civano has so far met the energy efficiency challenge. Civano was built to be a sustainable community in other

ways as well, by taking advantage of the area's abundant sunshine for water heating and for electricity generation, and by drastically reducing water use.

In order to reduce water use at Civano, the developers

1. Required the use of xeriscaping in all front yards. Xeriscape landscaping is suited to hot-dry climates and includes using native, drought-resistant plants that require a minimum of watering.
2. Made available reclaimed water and approved the use of water harvesting cisterns for landscape watering.
3. Required water-efficient appliances and fixtures in Civano homes.

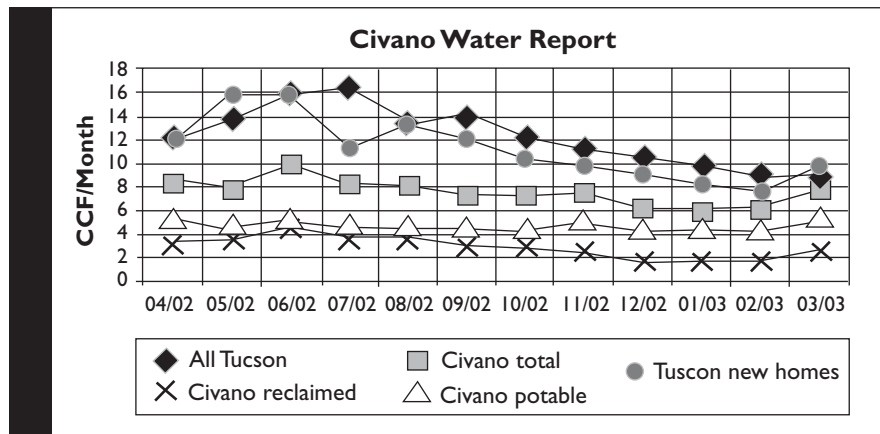
Civano Water Study

In 1998, Civano agreed to provide the city of Tucson with periodic updates

of its energy and water use to assess the compliance of homes with the SES. The engineering firm of Al Nichols, Incorporated, studied energy and water use at Civano. The firm compared the water use of 60 Civano homes built between 1997 and 2001 to the water use of 60 comparable homes in the greater

The total water use in new homes (310 gallons per day per house) compared to the water use in Tucson homes in general (335 gallons per day per house) suggests that new homes use approximately 7% less water per day due to contemporary building practices.

There are ten homes at Civano that don't use reclaimed water; instead, they use rainwater that is collected in cisterns for landscaping. Comparing these home's water use with that of Civano homes that use reclaimed water provided some insights. Homes that did not use reclaimed water used an average of 57 ft³



(left) **Figure 1.** Civano homes use about 60% of the water used by Tucson homes in general and 66% of the water used by new Tucson homes. (right) Some homes at Civano use rainwater collected in cisterns for landscaping.

Tucson area built between 1997 and 2001, and to the water use of Tucson homes that were built in any year.

The water study showed that Civano homes use an average of 125 gallons of potable water and 80 gallons of reclaimed water, or an average of 205 gallons total per day per house. This compares with an average of 335 gallons per day for Tucson homes in general, and 310 gallons per day for new Tucson homes. Civano homes use about 60% of the water used by Tucson homes in general and 66% of the water used by new Tucson homes. Civano homes use 62% less potable water than Tucson homes in general and 59% less potable water than new Tucson homes. (For a summary of water use comparisons by month, see Figure 1.)

The reduction in water use in Civano homes compared to Tucson homes is due to Civano's relatively small lot sizes; the use of reclaimed water and rainwater for landscaping needs; the practice of xeriscape landscaping; and water-saving measures within the homes, such as water-efficient dishwashers and clothes washers, and low-flow toilets and showerheads. The Nichols study showed that xeriscaping had the greatest positive impact on water savings.

Barriers to Doing Better

In an age of increasing concern over dwindling water supplies, especially in the West, the Civano water study is good news. There are now ways to drastically reduce water use inside and outside the home. And the data provided by the Nichols water study confirm which methods work well and which don't.

Although reclaimed water is not potable, using it for outdoor watering is an expensive strategy for an individual homeowner to use. The cost of an additional meter (potable and reclaimed water are metered separately), a backflow preventer, and expansion tanks adds considerably to the initial cost of homes that use reclaimed water. Furthermore, reclaimed water is more expensive on an ongoing basis. For most homes, potable water, provided by the Tucson Water Department, costs about \$1.03/ft³, while reclaimed water, provided by Pima County Wastewater Management, costs about \$1.31/ft³. And there is a higher than normal cost to maintain drip irrigation systems using reclaimed water, because of its relatively high mineral content.

per year of potable water (approximately 130 gallons per day per house), compared to 56 ft³ per year (approximately 128 gallons per day per house) for the homes that used reclaimed water. This represents a saving of only about 2%. Like all homes at Civano, these ten homes have small lots and use water efficiently. But for these homes, the startup costs for reclaiming water cannot be justified by the savings in the use of potable water.

New Civano homeowners currently have the option of not installing the reclaimed water system and installing rainwater-harvesting cisterns instead. This is a step in the direction of greater water savings for the community. But if the startup costs and cost of using reclaimed water aren't reduced, there will be no incentive for new Civano homeowners to sign up for reclaimed water service, and a chance for the community to save even more water will be lost.

—Jim Gunshinan

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For more information:

Go to www.civano.com.